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# Equilibrium Analysis of Energy Imbalance Reserve in Day-Ahead Electricity Markets

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***Abstract*—Energy imbalance reserve (EIR) product is a new product in the Independent System Operator of New England's (ISO-NE) day-ahead wholesale electricity market. Different from existing forward reserve products, EIR is a real option product, which is settled against the real-time energy price rather than reserve prices. This novel product has not been analyzed in the research literature in terms of its effects. In this paper, we develop a stochastic long-run equilibrium model that incorporates the risk preference of generator and demand agents participating in the energy and reserve market in both day-ahead and real-time. In a risk neutral environment, we find that the presence of the EIR product makes little difference on market outcomes. We also conduct a series of numerical simulations with risk-averse generators and demand, and observed increased advanced fuel procurement when the EIR product is present.**

***Index Terms*—Electricity markets, reserves, equilibrium analysis, complementarity problems, day-ahead markets, options**

## I. Background

New England's power is heavily reliant on natural gas-fired generating units. In fact, 55% of electricity in 2024 was produced from natural gas [1]. Natural gas is primarily delivered to New England via interstate pipelines, making it especially vulnerable during winter cold spells when residential heating demand surges. During cold spells, gas supply is often constrained and New England gas power plants that rely on just-in-time delivery of natural gas face challenges of procuring fuel in real time. These gas plants often lack incentives to secure gas in advance if they do not expect real-time operation (i.e., no day-ahead awards). Independent System Operator of New England (ISO-NE) has the second-lowest firm gas contracting rate among U.S. ISOs (with 45% in 2024) [2]. Instead, the gas powered fleet of ISO-NE opt for interruptible contracts on the spot market, which puts them at the lowest priority for fuel delivery when the natural gas infrastructure is constrained.

In addition, there is frequently a gap between the energy cleared in the day-ahead market (DAM) and the real-time load. This gap was positive (i.e. less energy cleared in the DAM than real time), in 64% of all hours in 2023, with an hourly average value of 515.7 MW and a maximum value of 3397.9 MW [3]. Similar persistent real time and day-ahead imbalances exist in other US market regions, even prompting the California system operator (CAISO) to create a new imbalance reserve product [4], [5]. Because additional resources needed to meet the gap are often procured after the DAM clears through the Reserve Adequacy Analysis (RAA) process, market clearing prices in the DAM do not reflect the actual cost of meeting real-time load, adversely affecting market transparency. More importantly, participants do not have strong incentives to procure fuel in advance if they do not have any DAM obligations.

In order to address these challenges [6], ISO-NE introduced an enhanced DAM on February 28th, 2025, including a novel ancillary service product called the Energy Imbalance Reserve option (EIR) [7]. EIR is designed to balance the gap between the real-time load predicted by ISO-NE and energy cleared in the DAM. The day-ahead market will now clear both day-ahead energy and energy imbalance reserve to meet the load forecast. EIR has an energy call "option" settlement where sellers receive an EIR payment in DAM and may incur a close-out cost based on real-time energy prices and a strike price set by the ISO.

One of ISO-NE's central objectives of introducing the EIR mechanism is to better align individual market participant's incentives with the system's overall reliability objectives. The ISO must ensure that sufficient generation is available to meet demand in all possible real-time scenarios. For a natural gas generator, this means it should secure adequate fuel supply ahead of the operating day instead of relying on the spot market in real time. However, purchasing fuel ahead of time exposes generators to the risk of financial loss if they are not dispatched, rendering fuel unused or resold. The goal of the EIR product is to mitigate this risk by providing a market-based mechanism where generators receive a day-ahead payment for EIR awards and face a real-time close-out cost if real-time energy prices exceed a pre-determined strike price.[1] By providing a financial signal ahead of real time, EIR may motivate generators to procure fuel when it is socially desirable but not privately profitable without the EIR mechanism. In addition, by incorporating the ISO's real-time load forecast as the forecast energy requirement (FER) constraint in the DAM and pricing it directly, the EIR product improves DAM price formation and reduces the need for out-of-market actions such as RAA. The goal of the FER is to secure sufficient energy and reserve from physical resources in the DAM to meet expected real time demand. Errors in the load forecast can adversely impact the day ahead energy market and reserve market. Underestimating real time load can result in insufficient generation committed and prepared

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[1]The strike price, determined daily by ISO-NE, is based on their forecast real time electricity price plus an additional ten dollars.

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to produce in real time, which can raise real time energy and reserve prices. Overestimating the load forecast would result in unnecessary levels of physical generation and reserve cleared in the market, relative to what is actually needed in real time, which result in higher costs to the consumer in the day ahead energy and reserve market.

## II. Modeling Framework, Literature Review and Contributions

EIR, a DAM-RTM balancing product with an option close-out, is unique from other reserves in different wholesale electricity markets. How the product may affect the decisions of different agents has not been studied in the literature. In this paper, we establish a framework for examining how the EIR option mechanism affects market participants behaviors and fuel procurement incentives. The models capture three types of agents, namely generators, demand and arbitragers, who participate in both the day-ahead and real-time markets under the uncertainty in real-time load and fuel prices.

Ancillary service markets around the United States include products designed to address real time uncertainty in the day ahead market associated with load forecast uncertainty and rapid changes to real time conditions.[2] Unlike typical contingency reserve products that target losses of elements in the bulk power system, these products aim to commit flexible generation capacity to provide the system with flexibility during volatile operational conditions. However, most of the existing products come under the umbrella, or are otherwise connected to ramping reserves. CAISO's Imbalance Reserve Product for instance was introduced to commit capacity day ahead to replace ramping reserves in real time. NYISO's Dyanmic Reserve Market Design (set to be implemented in 2027), contains a rule change that is most analogous to ISO-NE's EIR.[3]

The model follows a three-stage market structure, consistent with ISO's practices [8], [9], [10]. In the first stage (prior to the day-ahead energy market), physical generators choose tp commit to a discrete level of fuel investment to support real time production. In the second stage (day-ahead), DAM is resolved based on forecasted real-time conditions and day-ahead energy and EIR awards are determined. In the third stage (real-time), under each scenario representing a real-time demand and fuel price realization, generators make decisions on whether or not to purchase additional fuel on a spot market and on real time electricity generation (see Fig 1).

A key design objective of the EIR project is to encourage generators to undertake decisions ahead of real time product to improve their operational reliability. To capture this decision

[2]Note that these products only exist in the United States. Other restructured power markets around the world (NEM, NZ EM, EU Markets) lack markets products outside of the typical contingency reserves.

[3]It modifies the thirty minute reserve requirement to require the sum of cleared day ahead generation and reserves to be greater than or equal to the reserve requirement. Similar to EIR, the shadow price of the new constraint is an added premium to the day ahead price, and, when positive, increases the payments to all physical generation cleared in the day ahead market. Unlike EIR, it does not contain a real time call-option like settlement based on the real time price of electricity. Instead, it settles like a traditional operating reserve product, which is based on the real time reserve price.

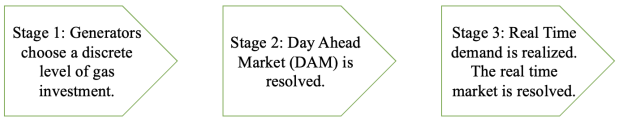



Fig. 1. Schematic of the three stages.

in our model, we model gas generators that participate in a simplified natural gas market with the decision to either purchase fuel ahead of production at a fixed price or to wait until real time to buy on a spot market where the price is scenario dependent. This approach illuminates if the presence of EIR incentivizes generators to procure fuel ahead of production. This would benefit reliability for the system, i.e. increase in securing fuel ahead of real time product, compared to the energy only market. We assume that any gas purchased at either stage must be used or subsequently sold at a cost lower than purchased (similar to a salvage value).

It is important to develop the multi-stage stochastic structure because market participants must make key decisions, such as day-ahead energy sales or purchases, EIR awards and day-ahead fuel purchase before the realization of uncertain real-time conditions. The stochastic nature of the model captures the uncertainty of possible real-time outcomes and enables us to evaluate risk-hedging behavior with and without the EIR product, in a so called steady state of the market.

In our framework, generators engage in a discrete-choice game in which each player selects a day-ahead fuel investment level from a finite set of feasible alternatives. Once these decisions are made, the corresponding advance gas procurement quantities, ($V_i^{DA}$), are fixed. The payoff for each resulting strategy profile is then computed from the competitive, risk-averse stochastic equilibrium of the subsequent two-stage market model. Hence, we model and solve a subgame perfect equilibrium problem. The payoffs for each agent in the finite game are a solution to a mixed complementarity problem (MCP) derived from the optimality conditions of the agents' profit maximization problems and market-clearing constraints with a fixed stage 1 gas investment level. To capture the impact of risk preference, the objective functions of participants' profit maximization problem use Conditional Value-at-Risk (CVAR). CVAR represents the generator's expected profits that fall below a percentile of the profit distribution.

The framework compares two market paradigms:

- Energy Market Only (EMO): A baseline energy only model without EIR that is a simplified reflection of ISO-NE's market prior to February 28, 2025
- Energy Market with Imbalance Reserve (EMIR): A co-optimized model where the EIR product is procured to meet a system-level forecast energy requirement (FER).

The use of mixed complementarity problems to model the equilibrium of multiple agents participating in an electricity market to study the effects of a particular policy is well established in the operations research literature on power markets. [11] models generators participating in both day-ahead and real time markets. [12] and [13] use two-stage stochastic equilibrium models to explore capacity expansion

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with risk-averse agents and utilize CVAR as their risk measure. [14] and [15] use MCPs to compare different market structures and policies, specifically how the presence of risk mitigation instruments changes investment behavior and social welfare. A comprehensive treatment of mixed complementarity modeling in energy systems is given in [16], with reviews of electricity market applications in [17], [18], and a tutorial on policy analysis in [19]. Outside of the MCP literature, [20] uses a bi-level optimization framework with risk-averse participants to evaluate the impact of virtual bidders in a renewable rich market environment and [21] looks at how multiple system risks can affect decision markers in power markets using a risk-factor oriented stochastic dominance approach. All the MCPs discussed in this paper were implemented in GAMS and solved using the PATH solver [22].

To maintain tractability and focus on impact of EIR production on equilibrium outcomes, we adopt the following modeling simplifications:

- Equilibrium models treat generators as fully dispatchable, omitting unit commitment and other intertemporal constraints. We follow the approximation commonly used in similar studies [11, 23, 24], as it is reasonable in large systems like ISO New England, which include hundreds of generators. Similarly, in [25], authors model UC, but relax the binary restriction and allow variables to take on any value between 0 and 1 to solve the model.
- Arbitragers are modeled as risk-neutral profit-maximizers with no transaction costs, a standard simplification in equilibrium studies [26] that facilitates analytical comparisons across market designs.
- The system is represented as a single node to focus on the core economic effects of EIR [13].
- The traditional operating reserve products, such as 10-minute and 30-minute reserves are not included in the model. The exclusion of these reserve products allows us to focus the model specifically on the EIR product and its intended function of incentivizing advance fuel procurement. While these operating reserve products are important for system reliability, their exclusion does not undermine the ability of the proposed equilibrium model to evaluate the specific role of EIR in influencing market clearing outcomes and fuel procurement behaviors.

Our modeling approach focuses on exploring two different aspects. First, we want to understand how the presence of energy imbalance reserve affects the solution of the equilibrium problems, specifically how it changes generators' fuel purchase decision and the behavior of a demand agent bidding in the day-ahead market. Secondly, the two sets of models, risk neutral and risk averse, shed some light in circumstances in which the introduction of EIRs may be an effective policy for achieving its design objectives.

The main contributions and findings of this paper are:

- This is the first work to develop an equilibrium model specifically for the evaluation of EIRs. The first stage is a discrete choice model designed to capture lumpiness of gas contracts. The generators payoffs for each set of choices are computed based on a competitive stochastic equilibrium that captures the vital demand uncertainty, and reflects a well regulated market such as ISO-NE.
- We show that when participants are risk neutral and generators' total capacity exceeds the forecasted real-time demand, the EIR product may not impact market outcomes or participants' fuel procurement behaviors relative to an energy-only DAM. This finding highlights that participant risk preferences play an important role in realizing the intended benefits of the EIR product.
- When participants are risk averse, it is shown that the presence of EIR can change generators' behaviors. In particular, generators become more willing to procure firm fuel in advance of operating day. The EIR mechanism helps shift risk-bearing from the system operator to risk-averse market participants.

The equilibrium modeling framework developed in this paper can be extended to analyze other market design questions related to uncertainty, fuel security, interactions between reliability and financial products, e.g. forward reserves.

TABLE I
SETS

| | |
|---|---|
| $i \in \mathcal{I}$ | Generators |
| $s \in \mathcal{S}$ | Real Time Demand Scenarios |

## III. Multi-Stage Stochastic Equilibrium Models for Markets With and Without EIR

In this section, we present three-stage stochastic competitive equilibrium models that characterize the behavior of three types of market participants, generators, demand and arbitragers, in both the conventional EMO and newly enhanced EMIR. In the first stage, generators select their gas contract amount, $NG_i^{DA}$, that represents the quantity of gas they will purchase on the day ahead market. For each agent type, we develop its profit maximization problem incorporating its risk preferences via CVAR, for the subsequent stages. These optimization problems are formulated separately for the market with energy only as well as the market with energy and EIR.

We compute a competitive equilibrium, by simultaneously solving all agents' optimization problems, along with market clearing conditions. Mathematically, this is achieved by converting each agent's optimization problem into its Karush-Kuhn-Tucker (KKT) conditions and stacking these KKT conditions (for all agents) along with market clearing conditions. This constitutes an MCP whose solutions correspond to equilibrium payoffs, given the generators' gas contract choice in stage one.

In a traditional electricity market, agents submit price–quantity bids to clear the market. In our model, agents choose only quantities, while prices are determined endogenously by the equilibrium to balance supply and demand across markets. Although this differs from the actual clearing process, it is appropriate when agents bid at true cost. This simplification does not affect the competitive equilibrium results, since price-taking agents respond to prices rather than influence them.

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TABLE II
VARIABLES

| | |
|---|---|
| $g_i^{DA}$ | Day-ahead Energy Award (MWh) |
| $e_i$ | Energy Imbalance Reserve (MWh) |
| $g_{i,s}^{RT}$ | Real Time Production (MWh) |
| $V_i^{DA}$ | Day-ahead Fuel Purchase (MWh) |
| $V_{i,s}^{RT}$ | Spot Fuel Purchase (MWh) |
| $W_{i,s}^{RT}$ | Fuel Sale (MWh) |
| $\lambda_s^{RT}$ | Real Time Energy Price ($/MWh) |
| $\lambda^{DA}$ | Day-ahead Energy Price ($/MWh) |
| $\rho$ | Energy Imbalance Reserve Price |
| $d^{DA}$ | Day-ahead Demand (MWh) |
| $\mu_{i,s}$ | Marginal Value of Fuel Investment ($/MWh) |
| $\eta$ | Value at Risk |
| $q_{i,s}$ | Risk Adjusted Probability |
| $Z_{i,s}$ | Per Scenario Profit of Generator i |
| $Z_s^D$ | Per Scenario Profit of Demand |

TABLE III
PARAMETERS

| | |
|---|---|
| $C_i$ | Marginal Production Cost Excluding Fuel Costs ($/MWh) |
| $C_{i,s}^I$ | Intraday Fuel Cost ($/MWh) |
| $C_i^F$ | Advanced Fuel Cost ($/MWh) |
| $R_{i,s}$ | Fuel Resale Price |
| $F_i$ | Generator Fuel Level (MWh) |
| $Q_i$ | Maximum Capacity (MWh) |
| $NG_i^{DA}$ | Gas Contract in Equivalent Power |
| $D_s^{RT}$ | Fixed Real Time Demand (MWh) |
| $D^{DA}$ | Fixed Day-ahead Demand (MWh) |
| $FER$ | Forecast Energy Requirement (MWh) |
| $\pi_s$ | Probability of Scenario s |
| $\alpha_i$ | Risk Level |
| $K$ | Strike Price ($/MWh) |

### A. Equilibrium Model for Energy Market Only

*1) Generators' Profit Maximization Problem in EMO:* Each generator maximizes risk-adjusted profit. Any real-time production must be backed by sufficient fuel, which can be procured either in advance (before the day-ahead energy market) or in real time. Generators may invest in advanced fuel equal to 0%, 25%, or 50% of capacity $Q_i$, represented by different $NG_i^{DA}$ values, at a fixed price $C_i^F$. This discrete choice captures the lumpy nature of long-term fuel contracting outside the spot market. Any additional real-time fuel is purchased at a stochastic spot price $C_{i,s}^I$, correlated with real-time demand scenario $s$. The generator's problem without EIR products is given below, with associated dual variables shown in brackets.

$$\max_{\Xi} \ \eta_i - \frac{1}{a_i}\sum_s \pi_s u_{i,s} \tag{1a}$$

$$\text{s/t} \ \ u_{i,s} \geq \eta_i - Z_{i,s} \qquad [q_{i,s}] \tag{1b}$$

$$g_i^{DA} \leq Q_i \qquad [\delta_i] \tag{1c}$$

$$g_{i,s}^{RT} \leq Q_i \qquad [\gamma_{i,s}] \tag{1d}$$

$$g_{i,s}^{RT} = F_i + V_i^{DA} + V_{i,s}^{RT} - W_{i,s}^{RT} \qquad [\mu_{i,s}] \tag{1e}$$

$$V_i^{DA} = NG_i^{DA} \tag{1f}$$

$$g_{i,s}^{RT}, g_i^{DA}, V_i^{DA}, V_{i,s}^{RT}, W_{i,s}^{RT}, u_{i,s} \geq 0 \qquad \forall s \tag{1g}$$

where

$$\Xi := (g_{i,s}^{RT}, g_i^{DA}, u_{i,s}, V_i^{DA}, V_{i,s}^{RT}, W_{i,s}^{RT}, \eta_i, u_{i,s})$$

$$Z_{i,s} := \lambda^{DA} g_i^{DA} + \lambda_s^{RT}(g_{i,s}^{RT} - g_i^{DA}) - C_i g_{i,s}^{RT} - C_i^F V_i^{DA} - C_{i,s}^I V_{i,s}^{RT} + R_{i,s} W_{i,s}^{RT} \tag{2}$$

The generator's objective function captures the conditional value at risk of its total day-ahead and real-time profit across uncertain real-time scenarios. This formulation follows the approach introduced by Rockafellar and Uryasev [27]. Each generator $i$ maximizes its risk adjusted profit by choosing a benchmark profit level $\eta_i$ and penalizing the expected profit below that level using. The non-negative variable $u_{i,s}$ reflects how much the actual profit in scenario $s$ is below this benchmark profit level $\eta_i$ via the first inequality constraint. I.e. the generator agent $i$ hedges against its lowest $\alpha_i$ quantile of profits. The parameter $\alpha_i \in (0, 1]$ corresponds to the generator's level of risk aversion. When $\alpha_i = 1$, the generator behaves in a risk-neutral manner and maximizes its expected profit. As $\alpha_i$ decreases, the generator becomes more risk-averse, emphasizing performance in the left tail of the profit distribution.

The second and third inequality constraints are related to physical limitations on day ahead awards and real time energy production, where $Q_i$ represents the economic maximum or capacity of a generator. The fourth equality constraint enforces $g_{i,s}^{RT}$ to be equal to some standard fuel level $F_i$, which can be thought of as a generator's starting fuel position before investing in additional fuel on the gas market and in this paper is set at a level of zero for all of the numerical examples, plus any day-ahead $V_i^{DA}$ or real time $V_{i,s}^{RT}$ fuel investment, and less any fuel leftover $W_{i,s}^{RT}$, which can be thought of as fuel to be sold after the real-time market is closed. $V_i^{DA}$ is set at one of the discrete choices of $NG_i^{DA}$ based on the generator's choice in the first stage. The corresponding shadow prices, $\mu_{i,s}$, can be interpreted as the marginal value of fuel investment.

*2) Demand's Profit Maximization Problem in EMO:* This section introduces a representative demand agent aggregating all demand-side participants, responsible for paying for all energy in EMO. The agent's sole decision variable is day-ahead demand $d^{DA}$; real-time demand is fixed and given by parameter $D_s^{RT}$. Demand agent's problem in absence of EIR options can be described as follows:

$$\max_{\eta^D, u_s^D, d^{DA}} \eta^D - \frac{1}{\alpha^D}\sum \pi_s u_s^D \tag{3a}$$

$$\text{s/t} u_s^D \geq \eta^D - Z_s^D \qquad [q_s^D] \tag{3b}$$

$$u_s^D, d^{DA} \geq 0 \qquad \forall s \tag{3c}$$

where

$$Z_s^D := -\lambda^{DA} d^{DA} - \lambda_s^{RT}(D_s^{RT} - d^{DA}) \tag{4}$$

The demand agent maximizes its risk adjusted profit, similar to the generator's counterpart. When $\alpha^D = 1$, the demand agent is risk neutral.

*3) Arbitragers' Profit Maximization Problem in EMO:* Every ISO market in the US allows for the participation of virtual market participants. These entities are strictly financial in that they neither produce nor consume electricity in real time, but merely seek to exploit arbitrage opportunities between the day-ahead and real time markets. They are able to offer in virtual supply (commonly referred to as incremental offers or

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INCs) or virtual demand (decremental bids or DECs), in the day-ahead market which is then closed out in real time. The arbitragers' problem is to maximize expected profits through their decision variables, $a$ and $b$, which respectively represent INCs and DECs.

$$\max_{a,b} \quad E_s[Z_s] \tag{5a}$$
$$\text{s/t} \quad a, b \geq 0 \tag{5b}$$
$$Z_s := (\lambda^{DA} - \lambda_s^{RT})a + (\lambda_s^{RT} - \lambda^{DA})b \tag{5c}$$

Arbitragers are assumed to be risk-neutral, reflecting the purely financial nature of their positions and their limited exposure to operational risk. They therefore maximize expected profit without penalizing rare adverse outcomes. In practice, small transaction or administrative costs (typically \$1–\$2/MWh) may deter arbitrage unless expected profits exceed this threshold, so day-ahead prices may not exactly equal expected real-time prices. For tractability, we assume costless arbitrage. This simplification isolates the effects of EIR and risk preferences on equilibrium outcomes and does not materially affect the qualitative insights, since such costs are very small relative to price levels and risk-driven market impacts.

*4) Market Clearing Conditions in EMO:* The day-ahead energy price is determined by the complementarity condition

$$\sum_i g_i^{DA} + a = d^{DA} + b \perp \lambda^{DA} \text{ free} \tag{6}$$

by ensuring total day-ahead energy supply and INCs equals total day-ahead bid-in demand and DECs. $\lambda^{DA}$ represents the marginal cost of supplying one more unit of energy in the day-ahead market. Because the condition is an equality, the associated price $\lambda^{DA}$ is unrestricted in sign. The real-time energy price is determined by the complementarity condition

$$\sum_i g_{i,s}^{RT} = D_s^{RT} \perp \lambda_s^{RT} \text{ free} \tag{7}$$

by ensuring total real-time energy production equals real-time materialized demand in each scenario. $\lambda_s^{RT}$ represents the marginal cost of supplying one more unit of energy in the real-time market under scenario $s$. Note that transmission constraints are ignored in the market clearing conditions for the tractability purpose. In practice, the New England system is not frequently congested and this simplification will not alter the main insights of the paper.

*5) MCP for EMO to Find the Subgame Equilibium Solution:* The full MCP for the EMO equilibrium, presented in Appendix B.1., is formed by appending the generators' optimality conditions to those of the arbitragers and demand agent, along with market clearing conditions requiring that supply equal demand in the real time and day-ahead markets. Any solution constitutes the equilibrium payoff for the specific choices of advanced gas contract investment by the generators.

*6) Discrete choice equilibrium:* To find a solution to the market equilibrium over three stages, we solve the MCP presented above, for every possible combination of agent investment decisions. Doing so provides $3^i$ different models with $i3^i$ risk-adjusted profits among all of the generators. A valid Nash equilibrium solution of the full game is a set of investment quantities such that each generator maximizes its own payoffs taking other generators' investment decisions as fixed. More formally, let $V_i$ be the set of all possible advanced fuel investments for generator i, let $v^* = (v_i^*, v_{-i}^*)$ be one particular investment scenario, where $v_i^*$ is the investment of player i, $v_{-i}^*$ is the investment of all other players, and let $K_i(v_i^*, v_{-i}^*)$ be the risk-adjusted profits of generator, which is equivalent to the objective function 1a, under one particular investment scenario. An equilibrium solution satisfies (8) for each generator $i$.

$$K_i(v_i^*, v_{-i}^*) \geq K_i(v_i, v_{-i}^*) \text{ for all } v_i \text{ in } V_i \tag{8}$$

### B. Equilibrium Model for Market with Imbalance Reserve

*1) Generators' Profit Maximization Problem in EMIR:* We similarly model the generator $i$'s optimization problem when EIR options are present.

$$\max_{\Xi} \quad \eta_i - \frac{1}{a_i}\sum_s \pi_s u_{i,s} \tag{9a}$$
$$\text{s/t} \quad u_{i,s} \geq \eta_i - Z_{i,s} \qquad [q_{i,s}] \tag{9b}$$
$$g_i^{DA} + e_i \leq Q_i \qquad [\delta_i] \tag{9c}$$
$$g_{i,s}^{RT} \leq Q_i \qquad [\gamma_{i,s}] \tag{9d}$$
$$g_{i,s}^{RT} = F_i + V_i^{DA} + V_{i,s}^{RT} - W_{i,s}^{RT} \qquad [\mu_{i,s}] \tag{9e}$$
$$V_i^{DA} = NG_i^{DA} \tag{9f}$$
$$g_{i,s}^{RT}, g_i^{DA}, e_i, V_i^{DA}, V_{i,s}^{RT}, W_{i,s}^{RT}, u_{i,s} \geq 0 \qquad \forall s \tag{9g}$$

where

$$\Xi := (g_{i,s}^{RT}, g_i^{DA}, u_{i,s}, V_i^{DA}, V_{i,s}^{RT}, W_{i,s}^{RT}, e_i, \eta_i)$$
$$Z_{i,s} := (\lambda^{DA} + \rho)g_i^{DA} + \lambda_s^{RT}(g_{i,s}^{RT} - g_i^{DA}) + \rho e_i$$
$$-[\lambda_s^{RT} - K]^+ e_i - C_i g_{i,s}^{RT} - C_i^F V_i^{DA} - C_{i,s}^I V_{i,s}^{RT} + R_{i,s} W_{i,s}^{RT}. \tag{10}$$

The main difference between the EMO and the EMIR models is that generators offer in EIRs at a price and quantity, much like how day-ahead energy offers are made (EIR being a financial product, we assume the generator does not incur any additional costs when providing EIR). In the day-ahead EMIR, a generator earns the EIR clearing price, $\rho$, for each MWh of reserve they provided, represented by $e_i$. The EIRs are purchased by the system at a strike price of $K$, which is determined (in advance and announced) by the system operator. In real-time any seller of EIR options, is charged $[\lambda_s^{RT} - K]^+$ for each MWh of EIR they have sold in the day-ahead. Note that the symbol + indicates there will be no cost to the generator provided $K$ is more than the real-time price. This real-time charge on the price of energy is rather unique, as reserves are commonly settled using reserve prices only.

Using $\lambda_s^{RT}$ in settling the EIR option implies that higher production in scenario $s$ results in lower closeout charges for reserves. In our risk-averse equilibrium models, this effect is manifested in the generator's behavior - specifically, their decision to invest in advance fuel as a hedge against potential closeout charges (see Section V.A for details). Under the EMIR

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mechanism, the day-ahead energy award $g_i^{DA}$ is not only compensated at the day-ahead energy clearing price $\lambda^{DA}$, but also at $\rho$ for its contribution to stratifying the FER as shown in the market clearing condition (13).

*2) Demand's Profit Maximization Problem in EMIR:* Similar to the EMO model, we model the demand agent represents the aggregate of all demand side market participants and pays for all energy and reserve. $d^{DA}$ is the sole decision variable of this agent, as we assume the real time loads are fixed. Demand agent's problem in the presence of EIR options follows.

$$\max_{\eta^D, u_s^D, d^{DA}} \quad \eta^D - \frac{1}{\alpha^D}\sum \pi_s u_s^D \tag{11a}$$

$$\text{s/t} \quad u_s^D \geq \eta^D - Z_s^D[q_s^D] \tag{11b}$$

$$u_s^D, d^{DA} \geq 0 \tag{11c}$$

where

$$Z_s^D := -(\lambda^{DA})d^{DA} - \lambda_s^{RT}(D_s^{RT} - d^{DA}) \\ -\rho(FER) + [\lambda_s^{RT} - K]^+ \sum_i e_i \tag{12}$$

*3) Arbitragers' Profit Maximization Problem in EMIR:* The arbitragers' profit maximization problem in EMIR is identical to the EMO counterpart described in subsection III-A3.

*4) Market Clearing Conditions in EMIR:* The day-ahead and real-time energy price complementarity conditions (6) and (7) remain in tact in EMIR model. In addition, price for EIR, $\rho$ is determined by the complementarity condition

$$0 \leq \sum_i g_i^{DA} + \sum_i e_i - FER \perp \rho \geq 0 \tag{13}$$

which stipulates that the EIR clearing price is positive only when FER equals the day-ahead energy awards and EIR awards.

*5) MCP for EMIR and Solution to the Full Game:* The full MCP for the EMIR subgame equilibrium is formed by the optimality conditions of the generator, demand, arbitragers, and market clearing conditions and is presented in Appendix B.2. Finding a valid equilibrium solution for the three stage game is done in the same manner as described in Section III.A.5.

## IV. Theoretical Results of the Risk Neutral Case

In the following section, we state theoretical findings for the risk neutral mixed complementarity models of the subgame. These models are special cases of the risk models when $\alpha_i = 1$, which represents agents maximizing their expected profits. Propositions IV.1, IV.3 and Corollary IV.1 refer to the risk neutral model with a demand agent, shown in the Appendix A1 and A2. We note that while our results pertain to models with a discrete choice of $V_i^{DA}$, the proofs trivially extend to the case where $V_i^{DA}$ is a continuous variable and is chosen to optimize a generator's profit concurrently with other variables, i.e. the risk neutral results remain valid for a continuous competitive two stage game.

**Proposition IV.1.** *Suppose that total available capacity is greater than $FER$ (i.e. $\sum_i Q_i > FER$). Then any valid solution of the MCP representing the Nash equilibrium of an EMIR subgame (for fixed choices of gas investment) satisfies $\rho = 0$. Furthermore, when the close out cost is not identically zero across all scenarios, $e_i = 0$ in equilibrium.*

*Proof.* Observe from (35) and (36) that the presence of arbitragers results in $\lambda^{DA} = E_s[\lambda_s^{RT}]$. Furthermore, since $\sum_i Q_i > FER$, either $\sum_i g_i^{DA} + \sum_i e_i > FER$ in which case $\rho = 0$ and we are done, or it must be that for some $\hat{i}$, $g_{\hat{i}}^{DA} + e_{\hat{i}} < Q_{\hat{i}}$. Therefore, from (33), we must have $\delta_{\hat{i}} = 0$. Now, (32) (together with the arbitrage condition) implies that $\rho \leq \delta_i$ for any $i$, and in particular, considering $\hat{i}$, this yields $\rho \leq 0$. Together with the right-hand-side of (39) this proves the result.

We make a further observation that if $E([\lambda_s^{RT} - K]^+) > 0$, that is, if the close out cost of the option is not identically zero in *all* scenarios, $e_i = 0$. This is follows directly from (34) together with $\delta_i \geq 0$ from (33) and the fact that $\rho$ must be zero as proved above. □

**Corollary IV.1.** *Under the same condition as Proposition IV.1, any solution of the EMIR MCP can be transformed to the corresponding equilibrium of the EMO MCP (with the corresponding choice of advanced fuel contract levels $NG_i^{DA}$). The spot fuel purchases and the total day-ahead energy values will be identical for EMIR and EMO MCP solutions.*

*Proof.* Note that in the MCP systems, conditions (15–19) are identical to (27–31), conditions (22–25) are identical to (35–38), and lastly (26) is identical to (40). Any difference in the two market schemes is attributed to the existence of $e_i$ and $\rho$. Now, let us denote the solutions to the EMIR MCP by overline. Given a solution to the market with EIRs, construct a solution to the market without EIRs as follows:

- Keep all real time values the same, e.g. $g_{i,s}^{RT} = \overline{g_{i,s}^{RT}}$, $V_{i,s}^{RT} = \overline{V_{i,s}^{RT}}$, $\lambda_s^{RT} = \overline{\lambda_s^{RT}}$, etc.
- For the remaining decision variables:

$$g_i^{DA} = e_i + \overline{g_i^{DA}},$$
$$b = \sum_i e_i + \overline{b},$$
$$\lambda^{DA} = \overline{\lambda^{DA}},$$
$$\delta_i = \overline{\delta_i}.$$

It is easy to see the above provides a solution satisfying (15–26). This completes the proof. □

**Proposition IV.2.** *Any subgame perfect equilibrium of the EMIR model is also a subgame perfect equilibrium of the EMO model.*

*Proof.* Following Corellary IV.1, an equilibrium solution, $v^*$ satisfying (8) is the same for the EIR case as the EMO case because the payoffs $K_i(v_i^*, v_{-i}^*)$ are the same for both markets. □

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**Remark**: If $g_i^{DA}$ in the solution of EMO MCP satisfy $\sum_i g_i^{DA} \geq FER$, we can construct a solution for the EMIR MCP by the following simple transformation:

$$\begin{aligned}\overline{g_{i,s}^{RT}} &= g_{i,s}^{RT},\\ \overline{\lambda_s^{RT}} &= \lambda_s^{RT}\\ \overline{g_i^{DA}} &= g_i^{DA},\\ \overline{\lambda^{DA}} &= \lambda^{DA},\\ \rho &= 0,\\ e_i &= 0.\end{aligned}$$

However, if $\sum_i g_i^{DA} < FER$ the natural transformation above will not render a solution for EMIR MCP. Note that any solution to the EMIR MCP will guarantee that $\sum_i g_i^{DA} \geq FER$ by its nature, as per Proposition IV.1.

**Proposition IV.3.** *In each of the EMIR and EMO MCPs described above, as long as $\lambda_s^{RT}$ are unique, we obtain unique day-ahead prices. The day-ahead and energy imbalance reserve quantities for an equilibrium solution are however not unique.*

*Proof.* Note that the arbitrage conditions enforce $\lambda^{DA} = E_s[\lambda_s^{RT}]$. As long as we have unique real time prices, all MCP solutions will exhibit uniqueness of day-ahead (and clearly real time) prices. However, the non-uniqueness of the equilibria corresponds to the (infinite) continuum of eligible $b$ and $g_i^{DA}$ that satisfy $\sum_i g_i^{DA} + a = d^{DA} + b$. □

## V. Observations and Numerical Examples

This section presents numerical simulation results that highlight the differences between the EMO and EMIR models when generators are risk averse. We begin with a simple case involving two generators to illustrate the underlying reasons for the models' distinct outcomes. We then extend the analysis to a much larger example drawn from ISO-NE data, with ten generators and ten real-time scenarios calibrated on the New England system. The purpose of these numerical experiments is to focus on how and why the introduction of EIR changes market outcomes when the generators are no longer risk neutral.[4]

### A. Two Generators

Consider a system with two competitive generators and five real time scenarios, its input parameters are presented in Tables IV and V. These numerical results compare generators' fuel procurement decisions across market structures under varying risk attitudes and $K$ values.

In the risk neutral case both the EMO and EMIR have identical payoffs and equilibrium solutions as proven in Section IV. Table VI shows the payoff matrix of the two generators under risk neutrality, in both the EMO and EMIR models. The rows of the matrix corresponds to Generator 2's advanced fuel investment strategies, while the columns capture Generator 1's different investment levels. The paired entries in the last three rows and columns of the table represent the payoffs of the generators with the first generator's payoffs coming first. Note that risk neutral generators do not invest in advanced fuel and purchase exclusively on the spot market. This is because advanced fuel cost exceeds the expected intraday cost. This observation persists in the EMO even when the generators are risk averse. Table VII shows results in a market with energy only under a range of $\alpha_i$ values. The generators do not invest in any advance fuel for any of the risk levels shown.

TABLE IV
Two Generator Inputs

| Scenario | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $D_s^{RT}$ | 50 | 75 | 100 | 125 | 150 |
| $C_{i,s}^{I}$ | 15 | 20 | 30 | 50 | 100 |
| $R_{i,s}$ | 0 | 0 | 0 | 0 | 0 |
| $\pi_s$ | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |

TABLE V
Two Generator Inputs

| | $Q_i$ | $C_i$ | $C_i^F$ |
|---|---|---|---|
| Gen 1 | 100 | 0 | 50 |
| Gen 2 | 100 | 5 | 50 |

However, in the EMIR case, risk averse generators become more willing to invest in advanced fuel, placing greater weight on adverse scenarios where intraday fuel costs and reserve closeout charges are elevated. The generators by investing in the more expensive fuel option for scenarios 1-3 are effectively sacrificing potential upside in more favorable scenarios to eliminate the possibility of incurring losses in scenarios 4 and 5. Advanced fuel investment acts as a risk mitigation strategy where it significantly reduces the impact of incurring high intraday fuel costs and penalties for deviations from their day-ahead energy position.

The fundamental difference in the behavior of the generators in the two different market paradigms is due to the presence of forecast energy requirement, which in the EMIR model forces the generators to provide either energy imbalance reserve or day-ahead energy. The generators must in some way participate in the day-ahead energy and reserve market, which creates greater variability in its profits across the different scenarios. In scenarios where fuel costs, real time deviations, and closeout costs are low, the generators realize positive profits, but when those same costs are higher it incurs losses. For risk-averse generators, the primary way to reduce losses in the unfavorable scenarios is to invest in advanced fuel, which provides a hedge by lowering fuel costs in scenarios when intraday costs exceed the fixed cost of advanced fuel. This investment smooths profits across scenarios and reduces the generators' exposure to downside risk. Thus, risk aversion drives greater fuel investment in the EMIR model compared to the EMO model, where no forward commitment is enforced and virtual supply can satisfy day ahead demand.

The observation that advanced fuel investment is greater in the EMIR model is not universally true, it critically depends on the strike price. For example, take generators under a range

[4]The following relatively small examples abstract away some of the complexity of the New England bulk power system, which contains over four hundred dispatchable generators. Fully representing this system would be computationally challenging and, more importantly, would obscure rather than clarify the economic mechanisms of interest.

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TABLE VI
TWO RISK NEUTRAL GENERATORS PAYOFF MATRIX

| | $V_i^{DA}$ | G1 | | |
|---|---|---|---|---|
| | | 0 | $0.25Q$ | $0.50Q$ |
| G2 | 0 | **(250,0)** | (109,0) | (-30,0) |
| | $0.25Q$ | (200,-200) | (60,-200) | (-160,-240) |
| | $0.50Q$ | (100,-440) | (-80,-520) | (-380,-640) |

TABLE VII
EMO ADVANCED FUEL INVESTMENTS VS $\alpha_i$

| $\alpha_i$ | 1 | .9 | .8 | .7 | .6 |
|---|---|---|---|---|---|
| v* | [0,0] | [0,0] | [0,0] | [0,0] | [0,0] |

of risk levels and four different strike prices, K = \$35, \$45, \$55, and \$65. The fuel investment quantities are shown in Table VIII. As the strike price increases and generators are sufficiently risk averse, total fuel investment decreases. This is because the source of its risk, the option closeout cost, is decreasing as the strike price grows. Additionally, there is a preference for the generators to provide energy imbalance reserve instead of day-ahead energy. As K increases, the generators' benefit of providing day-ahead energy over reserve decreases. Consider how much a generator would earn from choosing to provide day-ahead energy instead of reserve. It earns the price of energy and reserve and faces a deviation cost on each unit of day-ahead energy, $\lambda^{DA}+\rho-\lambda_s^{RT}$, and forgoes the reserve clearing price and closeout charge, $\rho-[\lambda_s^{RT}-K]^+$. When the closeout charge is positive, the costs of providing energy over reserve reduce to $\lambda^{DA}-K$, so as K increases above the day-ahead price, one can expect to see the overall amount of EIR awarded rise and day-ahead energy cleared fall. We state this more precisely below.

**Observation:** Note that the EIR mechanism provides a bona fide mechanism to mitigate risk and improve the generator's risk adjusted position. In particular, we can compare the profits attached to the scenarios of importance under risk aversion. For such a scenario $s$, the profit received by the generator in the EMO context is given by (14)

$$Z_{i,s} = \lambda^{DA} g_i^{DA} + \lambda_s^{RT}(g_{i,s}^{RT} - g_i^{DA}) - C_i g_{i,s}^{RT} - C_i^F V_i^{DA} - C_{i,s}^I V_{i,s}^{RT} \quad (14)$$

The same choices of decision variables would result in a feasible solution for the EMIR model. However, note that given the possibility of investing in EIRs, a generator can offer (and be dispatched) $e_i = \epsilon$ for EIRs and reduce their day-ahead generation to $g_i^{DA}-\epsilon$. For a high enough value of $K$, and small enough $\epsilon$, a generator would have an incentive to make this change. Consider the day-ahead profits (in scenario $s$). This would change from $(\lambda^{DA}-\lambda^{RT})g_i^{DA}$ to $(\lambda^{DA}-\lambda^{RT})g_i^{DA}+\rho g_i^{DA}-(\lambda^{DA}-\lambda_s^{RT}-\rho-[\lambda_s^{RT}-K]^+)\epsilon$.

TABLE VIII
ADVANCED FUEL INVESTMENT VS K AND $\alpha_i$

| | | $\alpha_i$ | | | | |
|---|---|---|---|---|---|---|
| | | 1 | .9 | .8 | .7 | .6 |
| K | 35 | [0,0] | [0,0] | [0,.5Q] | [.5Q,.25Q],[.25Q,.5Q] | [.5Q,.5Q] |
| | 45 | [0,0] | [0,0] | [0,.5Q] | [.5Q,.25Q] | [.5Q,.5Q] |
| | 55 | [0,0] | [0,0] | [0,.5Q] | [.25Q,.25Q],[0,.5Q] | [0,.5Q] |
| | 65 | [0,0] | [0,0] | [0,0] | [0,0] | [0,0] |

(Note that the real time profits would remain the same). Clearly for a proper choice of $\epsilon > 0$, the latter profits are greater than the former hence a generator will choose to sell EIRs and improve their risk adjusted profits.

### *B. Larger Numerical Example*

In this section, we present a larger scale numerical example calibrated on the ISO New England System. Using 2020-2024 datasets from the ISO New England's Capacity, Energy, Load, and Transmission Report (CELT) report, demand data, generator specific fuel consumption data from EIA-923, and the historical electric power price from EIR, we construct an example with ten scenarios and ten generators [28],[29],[2].Furthermore, our model includes a demand agent and two arbitraging agents. Arbitraging agents, which were not included in the previous section, assist in price discovery and eliminate differences between prices in the forward and real time market. In reality, there are numerous financial firms that participate in the New England wholesale market by providing price-quantity bids in the day ahead market that they must close out in real time. In the following example, there are two arbitrageurs, one for virtual demand and one for virtual supply, that buy or sell quantities only in the DAM. The results we present in this section are based on the continuous investment case as solving for fixed investment equilibria, for a large number of generators is computationally prohibitive (in particular, for this example, we would have to solve $3^{10}$ MCPs to find equilibria in the discrete case. The continuous investment case removes (1f) and (9f) from the respective EMO and EMIR generator optimization problems, eliminating the first stage discrete investment decision, The model with continuous investment becomes a perfectly competitive two-stage stochastic equilibrium model.

Due to the dominance of natural gas in New England and the focus of this paper, the assets represented in our example primarily use natural gas as their fuel type. Also according to [30], the price of natural gas in New England is more volatile relative to other fuel types such as oil, distillate, etc, hence generators reliant on alternative fuels are as less likely to be concerned with the choice of participation on the spot market as opposed to taking longer term forward positions in advance of production.

In the New England power system, there are forty different individual assets with natural gas as their primary fuel type corresponding to 21 different unique plants, which we aggregate at the level of market participant. We also assume a base load generation, with a capacity of 5 GW, that represents cheap non-gas generation (i.e. nuclear or renewable) that fully clears into the day ahead and real time markets and a peaker generator that effectively acts as a price cap on real time energy with a cost of generation equal to \$250/MWh.

In order to estimate the intraday fuel costs, we first take total fuel consumption and electricity generation from the EIA-923 dataset to calculate the heat rates for each individual plant. Using the heat rates and the state electric power plant price from EIA, we calculate the \$/MWh cost for each plant and then calculate the market participant's fuel costs as a weighted

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TABLE IX
SYSTEM INPUTS INCLUDING GENERATOR PARAMETERS AND SCENARIO-DEPENDENT DEMAND

| Component | Fixed Inputs | | | Intraday Fuel Cost $C^I_{i,s}$ | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $Q_i$ | $C^A_i$ | $C_i$ | **S1** | **S2** | **S3** | **S4** | **S5** | **S6** | **S7** | **S8** | **S9** | **S10** |
| *Generator Inputs* | | | | | | | | | | | | | |
| Gen 1 | 1381.2 | 60.7 | 0 | 40.39 | 41.32 | 43.22 | 43.78 | 45.53 | 47.20 | 47.50 | 59.92 | 67.23 | 70.76 |
| Gen 2 | 592.644 | 66.5 | 0 | 40.99 | 46.40 | 49.12 | 49.69 | 52.55 | 55.67 | 56.98 | 66.77 | 72.25 | 74.48 |
| Gen 3 | 3272.5 | 76.3 | 0 | 49.37 | 51.79 | 58.07 | 60.70 | 63.98 | 67.09 | 69.18 | 75.18 | 79.89 | 87.74 |
| Gen 4 | 1434.3 | 82.8 | 0 | 55.49 | 56.65 | 58.10 | 61.31 | 66.24 | 71.79 | 76.91 | 79.88 | 93.29 | 108.13 |
| Gen 5 | 317.1 | 91.1 | 0 | 60.58 | 67.95 | 70.82 | 71.54 | 73.83 | 77.28 | 81.49 | 86.27 | 102.35 | 119.16 |
| Gen 6 | 716.5 | 92.5 | 0 | 61.59 | 67.40 | 73.39 | 75.01 | 78.12 | 82.94 | 89.69 | 93.59 | 97.19 | 105.68 |
| Gen 7 | 1343.6 | 99.0 | 0 | 59.63 | 61.45 | 65.10 | 72.14 | 79.71 | 90.53 | 107.33 | 115.02 | 117.77 | 121.75 |
| Gen 8 | 608.3 | 115.6 | 0 | 86.60 | 89.62 | 94.04 | 98.21 | 101.55 | 104.32 | 106.79 | 110.47 | 124.82 | 139.40 |
| *Real Time Demand and Probability* | | | | | | | | | | | | | |
| $D^{RT}_s$ | – | – | – | 12038 | 12362 | 12673 | 12980 | 13271 | 13539 | 13807 | 14069 | 14354 | 14620 |
| $\pi_s$ | – | – | – | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 | .1 |

average of the different plants in its operation. Additionally we use the assumption that the only cost incurred by the natural gas generators in order to produce electricity in real time is fuel purchase, so $C_i$ = 0. The cost of intraday fuel for each individual generator in each individual scenario is a function of their assumed heat rate and the scenario dependent price of natural gas. The real time demand scenarios are constructed based on hourly system load in New England, and it takes on a range from 10.6 GW to 13.3 GW. The price of natural gas is directly correlated with the price of electricity, so as real time demand increases so does $C^I_{i,s}$. Unless stated otherwise, the advanced fuel cost for each natural gas generator, $C^F_i$ , is the expected intraday fuel cost for each agent plus an additional \$10/MWh. FER is equal to 8.5 GW, which is $E[D^{RT}_s] - BaseloadGen + 500$. Lastly, K is set to the expected real time price plus ten dollars, which is how the strike price is set in ISO New England. The input parameters are presented in Tables IX. The purpose of these numerical results, similar to the previous section, is to explore the differences in solutions under different market structures and input parameters. We begin our presentation of numerical results with a comparison between market outcomes in the EMO and EMIR market under three different cases. Each case corresponds to different assumed risk levels for Generators 1-5, all other input parameters are identical. The remaining generators, demand agent, and virtuals are risk-neutral. The cases are as follows:

- Case 1: $\alpha_i = 1\ \forall i$.
- Case 2: $\alpha_i = 0.5$ for $i = \{1, 2, 3, 4, 5\}$.
- Case 3: $\alpha_i = 0.25$ for $i = \{1, 2, 3, 4, 5\}$.
- Case 4: $\alpha_i = 0.25$ for $i = \{1, 2\}$ and $\alpha_i = 0.5$ for $i = [3, 4, 5]$

Selected outcomes under the four different cases are presented in Table X. In the case when all market participants are risk neutral, there is very little difference between the two markets, as indicated earlier by Corollary IV.1 and Proposition IV.1. The profits of the different generators, the fuel procurement levels, prices in DA/RT markets, and consumer costs are identical.

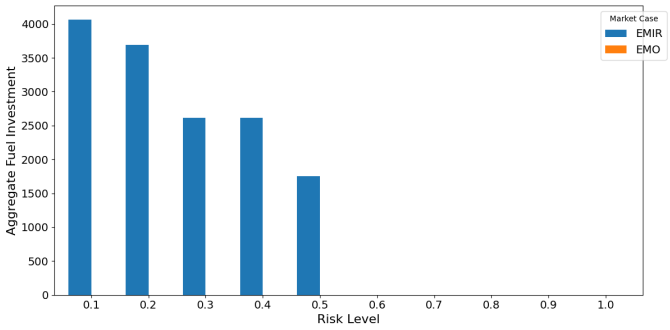



Fig. 2. Fuel Investment vs Risk

The sole difference between the EMO and EMIR is quantities traded by the different agents in the day ahead market. In the EMIR, physical generation clears 8500 MWh in the DA market, whereas in the EMO they receive no DA awards. Virtual trading and physical demand quantities are different in the risk neutral case, but these differences make no impact on the agents' profits or market prices.

In Case 2, differences between the EMO and EMIR models begin to emerge. Advanced fuel investment across the risk averse agents increases in the EMIR case to 1751.8 MWh, while they stay at 0 in the EMO model. Also, a positive EIR price emerges, which substantially increases the total consumer costs from \$787,600 to \$943,800, while the costs to consumer remains constant in the EMO across the three cases. This increase results from the the EIR price $\rho$ now being a positive markup on the day ahead price as well as the pure EIR price. The increase in payments to generators with a day ahead award is the reason consumer costs spike.

In Case 3, we observe similar changes, with advanced fuel cost, consumer cost, and EIR prices increasing in the EMIR model, and little to no changes in the EMO model, outside of risk adjusted profits for the risk-averse generators. In both the EMO and EMIR, the expected real time price and day ahead prices stay constant across all cases tested. The increasing risk-aversion of the particular generators causes a higher $\rho$, which raises the consumer costs.

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TABLE X
COMPARISON OF MARKET OUTCOMES FOR SELECT GENERATORS, CONSUMER, AND VIRTUALS FOR EMIR AND EMO

| Performance Metric | EMO | | | | EMIR | | | |
|---|---|---|---|---|---|---|---|---|
| | **Case 1** *(Risk Neutral)* | **Case 2** *(Rel. RA)* | **Case 3** *(Very RA)* | **Case 4** *(Asym. RA)* | **Case 1** *(Risk Neutral)* | **Case 2** *(Rel. RA)* | **Case 3** *(Very RA)* | **Case 4** *(Asym. RA)* |
| ***Risk-Adjusted Profits ($ Thousands) and Expected Consumer Costs*** | | | | | | | | |
| Gen 1 | 56.4 | 53.1 | 52.3 | 52.3 | 56.4 | 71.5 | 70.3 | 68.4 |
| Gen 2 | 20.8 | 19.7 | 19.2 | 19.2 | 20.8 | 27.0 | 26.8 | 26.3 |
| Gen 3 | 82.5 | 75.9 | 74.8 | 74.8 | 82.5 | 114.9 | 115.5 | 115.7 |
| Gen 4 | 26.9 | 23.5 | 21.5 | 23.5 | 26.9 | 38.9 | 41.3 | 39.3 |
| Gen 5 | 3.71 | 2.59 | 2.11 | 2.59 | 3.71 | 5.95 | 6.49 | 6.05 |
| Total Consumer Costs | 787.6 | 787.6 | 787.6 | 787.6 | 787.6 | 943.8 | 958.5 | 946.5 |
| ***Day Ahead Market Awards (MWh)*** | | | | | | | | |
| DA Awards (Supply) | 0 | 6434.1 | 4067.7 | 3940.0 | 8500 | 8500 | 8500 | 8500 |
| Virtual Supply | 8866.8 | 4799.0 | 2467.8 | 5214.1 | 0 | 15.3 | 0 | 0 |
| DA Demand (Physical) | 0 | 0 | 35.1 | 287.3 | 4250 | 4257.6 | 0 | 0 |
| Virtual Demand | 8866.8 | 8866.8 | 8866.8 | 8866.8 | 4250 | 4257.6 | 8500 | 8500 |
| EIR (Physical) | – | – | – | – | 0 | 0 | 0 | 0 |
| ***Fuel Procurement (MWh)*** | | | | | | | | |
| Gen 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1381.2 | 443.7 |
| Gen 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Gen 3 | 0 | 0 | 0 | 0 | 0 | 0 | 626.7 | 0 |
| Gen 4 | 0 | 0 | 0 | 0 | 0 | 1434.3 | 1434.3 | 1434.3 |
| Gen 5 | 0 | 0 | 0 | 0 | 0 | 317.5 | 317.5 | 317.5 |
| ***Market Prices ($/MWh)*** | | | | | | | | |
| Expected Real Time Price | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 |
| DA Energy Price | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 | 91.5 |
| EIR Reserve Price | – | – | – | – | 0 | 18.38 | 20.1 | 18.7 |

*Note:* RA = Risk Averse. EMO market structure does not include an EIR reserve product.

Figure 2 shows advanced fuel investment for all generators under a range of $\alpha$ values. Similar to Cases 2 and 3 in this section, we only vary the risk attitudes of generators one through five, with the remaining generators remaining risk neutral. For $\alpha = (1, 0.6)$, EMO and EMIR advanced fuel investment are both zero. When the selected generators become more risk averse in the EMIR case, they purchase more advanced fuel with a maximum value of 4037 MW, which is about 45% of the natural gas capacity in our example and 32.5% of the highest real time demand scenario. Generators participating in a market with energy only invest in no advanced fuel for all $\alpha$ values tested. In the EMO case, generators are not required to participate in the day-ahead energy market, which reduces their exposure to risk, and allows them to achieve sufficient risk adjusted profits without investing in advanced fuel.

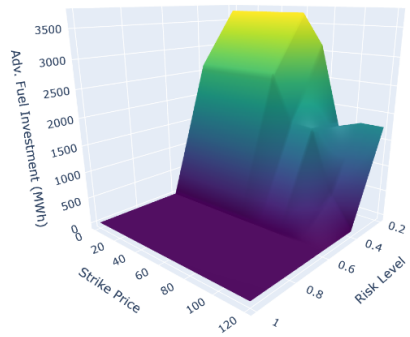



Fig. 3. Advanced Fuel Investment vs Risk and Strike Price without Virtuals and Fixed Demand

In the two generator example, fuel investment in EMIR was sensitive to the strike price. As $K$ increased, $\sum_i V_i^{DA}$ decreased, since lesser closeout costs diminish the incentive for risk averse generators to invest in advanced fuel (recall that high $K$ mitigated risk of a closeout cost in scenarios with both high $\lambda_s^{RT}$ and high spot fuel costs). However, in this large example with, a demand agent and virtual participants, advanced fuel investment is no longer sensitive to changes in $K$. Figure 4 shows $\sum_i V_i^{DA}$ under different strike prices and risk attitudes, with a clear positive relationship between $\alpha$ and fuel investment. However for any risk level, aggregate fuel investment remains unchanged for all $K$ values. We note that FER is entirely met by the procured day ahead energy, with no EIR traded. (In particular, we find that $b - a + d^{DA} = \sum_i g_i^{DA} = FER$, rendering $\sum_i e_i = 0$, and making $K$ irrelevant to generator decisions. When we eliminate virtual traders from the market and replace the demand agent with a fixed day ahead demand bid (assumed to be less than FER), then generators provide some EIR and $K$ becomes a relevant mechanism again. As shown in Figure 3, the pattern from the simpler numerical example reemerges, with $\sum_i V_i^{DA}$

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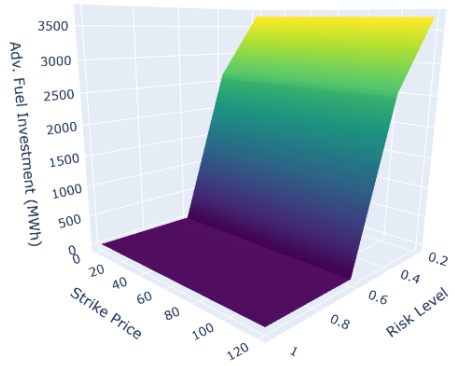



Fig. 4. Advanced Fuel Investment vs Risk and Strike Price with Virtuals and a Demand Agent

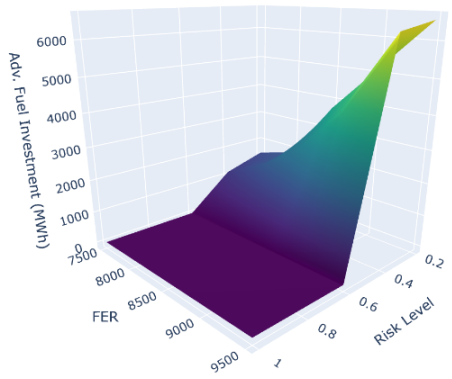



Fig. 5. FER vs Advanced Fuel Investment

increasing as $K$ decreases.

In addition to $K$, FER is an important aspect of the EIR market that impacts generator decision. Figure 5 plots $\alpha$, FER, and advanced fuel investment, and demonstrates that as FER increases so does advanced fuel investment when generators are sufficiently risk averse. Increasing FER also increases the costs to consumer, so the increased fuel investment does come at a cost to demand. So, while, the EIR product, assuming a careful choice of its associated elements K and FER, are effective at inducing a socially beneficial change in generators' behavior, this ultimately comes at a higher cost to demand.

The last parameter for our sensitivity analysis is the cost of advanced fuel. For all simulations, the cost of advanced fuel is set at the expected intraday fuel cost plus ten dollars. Figure 6 presents fuel investment for different risk levels $\alpha = (1, 0.8, 0.6, 0.4, 0.2)$ and advanced fuel costs. Fuel price adjustment is the premium above the expected intraday fuel cost, $E[C^I_{i,s}]$. In the EMIR case, we observe that fuel investment decreases as the advanced fuel cost increases, as even risk-averse generators become less willing to pay much higher relative amounts to decrease their fuel costs in particular scenarios. When sufficiently risk-averse, generators are still willing to invest in advanced fuel even when it is substantially more expensive than the expected spot price. In the EMO case, generators only invest in advanced fuel when they are risk averse and the price of day ahead fuel is either exactly equal to the expected spot price or the fuel price adjustment is \$5.

All previous results involved a risk-neutral demand agent. When the demand agent becomes risk-averse, it increases its day-ahead demand purchases above and beyond the load forecast in both the EMIR and EMO models. Across all models, physical demand purchases day-ahead energy in excess of the highest real time demand scenario at every $\alpha_i$ value below 1. In other words, a risk averse demand agent buys more energy on the DA market than it will consume in real time regardless of whether EIR is present or not.

| $\alpha$ | 1 | 0.8 | 0.6 | 0.4 | 0.2 |
|---|---|---|---|---|---|
| EMO | 0 | 11255 | 11027 | 11267 | 11379 |
| EMIR | 4250 | 11255 | 11027 | 11267 | 11379 |

TABLE XI
DAY AHEAD QUANTITIES PURCHASED BY RISK AVERSE DEMAND

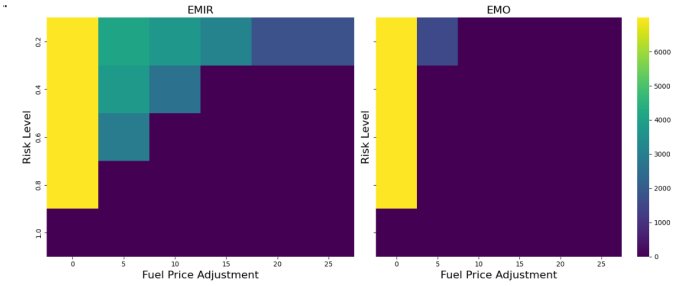



Fig. 6. Advanced Fuel Investment vs $C^{DA}_i$

## VI. DISCUSSION

In this paper, we introduced, modeled, and analyzed the potential effects of ISO New England's Energy Imbalance Reserve product by developing simple equilibrium models that incorporate risk-averse producers and consumers, both with and without the EIR mechanism. We model risk using conditional value at risk, a coherent risk measure with convexity assumptions. Similarly, our simplified models are convex, ignoring unit commitment and other intertemporal constraints. Our analysis revealed that under risk-neutral assumptions, outcomes in the two market designs—EMIR and EMO—were nearly indistinguishable. In the risk neutral case, the presence of the EIR had negligible impact on supply or demand side behavior, indicating that the reserve product's effectiveness depends on market participants' risk attitudes.

When risk aversion was introduced, especially on the supply side, notable differences emerged between the two paradigms. Risk-averse generators in the EMIR model were more inclined to undertake higher-cost investments in advanced fuel to mitigate their risk. These investments were particularly sensitive to a number of input parameters including the risk attitude of the generators, the forecast energy requirement, and, under certain conditions, the strike price parameter $K$. When generators opt to provide EIR, as $K$ increased, the incentive to invest in advanced fuel declined due to the lower option closeout charge. This trade-off underscores the importance of carefully choosing $K$ to balance incentives for both reliability and cost-effective reserve provision. When FER increased, so did the fuel investment of the generators, but it came at a greater cost to the consumer. The ISO must be careful in how they set

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the key elements of the EIR market, $K$ and $FER$, to ensure the system reliability benefits balance the costs to both the consumers and generators.

While these findings offer valuable insights, translating them directly into policy recommendations should be done with caution. Our theoretical results simplify power system operations as we do not consider unit commitment decisions or model intertemporal constraints. In the risk-averse case studies, the full complexity of the New England bulk power system is reduced to enable computation. The EIR market is already operational, and empirical evaluations by independent market monitors will be crucial in assessing its real-world impacts. Our study was forward-looking and aimed at providing a theoretical lens on possible behavioral shifts prior to market implementation. Nonetheless, two key takeaways emerge from our work. Future work could extend this analysis by incorporating additional features, such as strategic behavior by market participants or transmission constraints to see how a nodal vs system wide strike price can impact economic incentives. Empirical studies comparing real-world EIR performance against results of this paper will be instrumental in refining modeling approaches going forward. Lastly, the EIR product is one possible solution in enhancing real-time system reliability. Other market products or rules like CAISO's imbalance reserves or NYISO's dynamic reserves initiative are also aimed at correcting day and real time load imbalances. An interesting direction for future work could be a study that models and compares the efficacy of those different market products.

## VII. Acknowledgment

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## VIII. Appendix

### A. Risk Neutral Models

#### 1) Risk Neutral EMO Model:

$$0 \le C_i\pi_s + \mu_{i,s} - \lambda_s^{RT}\pi_s + \gamma_{i,s} \perp g_{i,s}^{RT} \ge 0 \tag{15}$$
$$0 \le Q_i - g_{i,s}^{RT} \perp \gamma_{i,s} \ge 0 \tag{16}$$
$$g_{i,s}^{RT} = F_i + V_{i,s}^{RT} + NG_i^{DA} - W_{i,s}^{RT} \perp \mu_{i,s} \text{ free} \tag{17}$$
$$0 \le C_{i,s}^{I}\pi_s - \mu_{i,s} \perp V_{i,s}^{RT} \ge 0 \tag{18}$$
$$0 \le \mu_{i,s} - R_{i,s}\pi_s \perp W_{i,s}^{RT} \ge 0 \tag{19}$$
$$0 \le \sum_s \pi_s\lambda_s^{RT} - \lambda^{DA} + \delta_i \perp g_i^{DA} \ge 0 \tag{20}$$
$$0 \le Q_i - g_i^{DA} \perp \delta_i \ge 0 \tag{21}$$
$$0 \le \sum_s \pi_s\lambda_s^{RT} - \lambda^{DA} \perp a \ge 0 \tag{22}$$
$$0 \le \lambda^{DA} - \sum_s \pi_s\lambda_s^{RT} \perp b \ge 0 \tag{23}$$
$$\sum_i g_i^{DA} + a = d^{DA} + b \perp \lambda^{DA} \text{ free} \tag{24}$$
$$\sum_i g_{i,s}^{RT} = D_s^{RT} \perp \lambda_s^{RT} \text{ free} \tag{25}$$
$$0 \le \lambda^{DA} - \sum_s \pi_s\lambda_s^{RT} \perp d^{DA} \ge 0 \tag{26}$$

#### 2) Risk Neutral EMIR Model:

$$0 \le C_i\pi_s + \mu_{i,s} - \lambda_s^{RT}\pi_s + \gamma_{i,s} \perp g_{i,s}^{RT} \ge 0 \tag{27}$$
$$0 \le Q_i - g_{i,s}^{RT} \perp \gamma_{i,s} \ge 0 \tag{28}$$
$$g_{i,s}^{RT} = F_i + NG_i^{DA} + V_{i,s}^{RT} - W_{i,s}^{RT} \perp \mu_{i,s} \text{ free} \tag{29}$$
$$0 \le C_{i,s}^{I}\pi_s - \mu_{i,s} \perp V_{i,s}^{RT} \ge 0 \tag{30}$$
$$0 \le \mu_{i,s} - R_{i,s}\pi_s \perp W_{i,s}^{RT} \ge 0 \tag{31}$$
$$0 \le \sum_s \pi_s\lambda_s^{RT} - \lambda^{DA} - \rho + \delta_i \perp g_i^{DA} \ge 0 \tag{32}$$
$$0 \le Q_i - g_i^{DA} - e_i \perp \delta_i \ge 0 \tag{33}$$
$$0 \le \sum_s \pi_s(\lambda_s^{RT} - K)^+ - \rho + \delta_i \perp e_i \ge 0 \tag{34}$$
$$0 \le \sum_s \pi_s\lambda_s^{RT} - \lambda^{DA} \perp a \ge 0 \tag{35}$$
$$0 \le \lambda^{DA} - \sum_s \pi_s\lambda_s^{RT} \perp b \ge 0 \tag{36}$$
$$\sum_i g_i^{DA} + a = d^{DA} + b \perp \lambda^{DA} \text{ free} \tag{37}$$
$$\sum_i g_{i,s}^{RT} = D_s^{RT} \perp \lambda_s^{RT} \text{ free} \tag{38}$$
$$0 \le \sum_i g_i^{DA} + \sum_i e_i - FER \perp \rho \ge 0 \tag{39}$$
$$0 \le \lambda^{DA} - \sum_s \pi_s\lambda_s^{RT} \perp d^{DA} \ge 0 \tag{40}$$

### B. Risk Averse Models

#### 1) EMO:

$$0 \le C_i q_{i,s} + \mu_{i,s} - \lambda_s^{RT} q_{i,s} + \gamma_{i,s} \perp g_{i,s}^{RT} \ge 0 \tag{41}$$
$$0 \le C_i^F - \sum_s \mu_{i,s} \perp V_i^{DA} \ge 0 \tag{42}$$
$$0 \le C_{i,s}^{I} q_{i,s} - \mu_{i,s} \perp V_{i,s}^{RT} \ge 0 \tag{43}$$
$$0 \le \mu_{i,s} - R_{i,s}^{RT} q_{i,s} \perp W_{i,s}^{RT} \ge 0 \tag{44}$$
$$0 \le \sum_s q_{i,s}\lambda_s^{RT} - \lambda^{DA} + \delta_i \perp g_i^{DA} \ge 0 \tag{45}$$
$$\sum_s q_{i,s} = 1 \perp \eta_i \text{ free} \tag{46}$$
$$0 \le \frac{1}{\alpha_i}\pi_s - q_{i,s} \perp u_{i,s} \ge 0 \tag{47}$$
$$0 \le Q_i - g_{i,s}^{RT} \perp \gamma_{i,s} \ge 0 \tag{48}$$
$$g_{i,s}^{RT} = F_i + V_{i,s}^{RT} + V_i^{DA} - W_{i,s}^{RT} \perp \mu_{i,s} \text{ free} \tag{49}$$
$$0 \le Q_i - g_i^{DA} \perp \delta_i \ge 0 \tag{50}$$
$$0 \le u_{i,s} - \eta_i + Z_{i,s} \perp q_{i,s} \ge 0 \tag{51}$$
$$0 \le \lambda^{DA} - \sum_s q_s^D\lambda_s^{RT} \perp d^{DA} \ge 0 \tag{52}$$
$$0 \le u_s^D - \eta^D + Z_s^D \perp q_s^D \ge 0 \tag{53}$$
$$\sum_s q_s^D = 1 \perp \eta^D \text{ free} \tag{54}$$
$$0 \le \frac{1}{\alpha^D}\pi_s - q_s^D \perp u_s^D \ge 0 \tag{55}$$
$$0 \le E_s[\lambda_s^{RT}] - \lambda^{DA} \perp a \ge 0 \tag{56}$$
$$0 \le \lambda^{DA} - E_s[\lambda_s^{RT}] \perp b \ge 0 \tag{57}$$
$$\sum_i g_i^{DA} + a = d^{DA} + b \perp \lambda^{DA} \text{ free} \tag{58}$$
$$\sum_i g_{i,s}^{RT} = D_s^{RT} \perp \lambda_s^{RT} \text{ free} \tag{59}$$
$$NG_i^{DA} = V_i^{DA} \tag{60}$$

#### 2) Risk Averse EMIR:

$$0 \le C_i q_{i,s} + \mu_{i,s} - \lambda_s^{RT} q_{i,s} + \gamma_{i,s} \perp g_{i,s}^{RT} \ge 0 \tag{61}$$
$$0 \le C_i^F - \sum_s \mu_{i,s} \perp V_i^{DA} \ge 0 \tag{62}$$
$$0 \le C_{i,s}^{I} q_{i,s} - \mu_{i,s} \perp V_{i,s}^{RT} \ge 0 \tag{63}$$
$$0 \le \mu_{i,s} - R_{i,s} q_{i,s} \perp W_{i,s}^{RT} \ge 0 \tag{64}$$
$$0 \le \sum_s q_{i,s}\lambda_s^{RT} - \lambda^{DA} - \rho + \delta_i \perp g_i^{DA} \ge 0 \tag{65}$$

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$$0 \leq \sum_s q_{i,s}[\lambda_s^{RT} - K]^+ - \rho + \delta_i \perp e_i \geq 0 \tag{66}$$

$$\sum_s q_{i,s} = 1 \perp \eta_i \text{ free} \tag{67}$$

$$0 \leq \frac{1}{\alpha_i}\pi_s - q_{i,s} \perp u_{i,s} \geq 0 \tag{68}$$

$$0 \leq Q_i - g_{i,s}^{RT} \perp \gamma_{i,s} \geq 0 \tag{69}$$

$$g_{i,s}^{RT} = F_i + V_{i,s}^{RT} + V_i^{DA} - W_{i,s}^{RT} \perp \mu_{i,s} \text{ free} \tag{70}$$

$$0 \leq Q_i - g_i^{DA} - e_i \perp \delta_i \geq 0 \tag{71}$$

$$0 \leq u_{i,s} - \eta_i + Z_{i,s} \perp q_{i,s} \geq 0 \tag{72}$$

$$0 \leq \lambda^{DA} - \sum_s q_s^D \lambda_s^{RT} \perp d^{DA} \geq 0 \tag{73}$$

$$0 \leq u_s^D - \eta^D + Z_s^D \perp q_s^D \geq 0 \tag{74}$$

$$\sum_s q_s^D = 1 \perp \eta^D \text{ free} \tag{75}$$

$$0 \leq \frac{1}{\alpha^D}\pi_s - q_s^D \perp u_s^D \geq 0 \tag{76}$$

$$0 \leq E_s[\lambda_s^{RT}] - \lambda^{DA} \perp a \geq 0 \tag{77}$$

$$0 \leq \lambda^{DA} - E_s[\lambda_s^{RT}] \perp b \geq 0 \tag{78}$$

$$\sum_i g_i^{DA} + a = d^{DA} + b \perp \lambda^{DA} \text{ free} \tag{79}$$

$$\sum_i g_{i,s}^{RT} = D_s^{RT} \perp \lambda_s^{RT} \text{ free} \tag{80}$$

$$0 \leq \sum_i g_i^{DA} + \sum_i e_i - FER \perp \rho \geq 0 \tag{81}$$

$$NG_i^{DA} = V_i^{DA} \tag{82}$$

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